Muhammad Memon

Graduate Computer Engineer in Mobile Systems

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Portfolio: Engineering Portfolio.web | GitHub: MMemon2003/README.md at main · MMemon2003/MMemon2003

PROFESSIONAL SUMMARY

Computer Engineering graduate (BSc Hons, 2:1) from TU Dublin with hands-on experience building automated monitoring and data-acquisition systems, from sensors and microcontrollers to Node-RED, MQTT and Grafana dashboards. Experienced in calibrating sensors, testing real-time performance, and methodically diagnosing faults, with a strong commitment to strict safety and procedural standards. Seeking a graduate automation engineering role in Dublin.

TECHNICAL SKILLS

• **Automation and Data Flow:** Node-RED, MQTT, IoT telemetry, real-time sensor data acquisition, InfluxDB, Grafana • **Embedded and Hardware:** ESP8266/ESP32, Raspberry Pi, microcontrollers, I2C, sensor calibration, PCB layout • **Programming:** C, C++, Python, Java, JavaScript, SQL

- **Testing and Validation:** Debugging, system validation, performance testing, latency testing
- **Systems and Cloud:** Microsoft Azure, Linux, systems administration, computer networks, database design
- **Professional:** Teamwork, problem solving, communication, time management

EDUCATION

Technological University Dublin, Blanchardstown | 2026

BSc (Hons) Computer Engineering (TU807), GPA: 3.0

Relevant modules: Software and Real-Time Embedded Systems, Computer Networks, Java and Algorithms

PROJECTS

Networked Microcontroller IoT System (3rd Year Project)

- Designed and deployed a real-time ESP8266 monitoring system using Azure and MQTT telemetry with Grafana dashboards.
- Diagnosed communication failures through structured debugging.

Greenhouse Monitoring System (2nd Year Project)

• Designed an embedded environmental monitor measuring temperature, humidity and light. • Calibrated sensors, improved PCB layout for signal integrity and ran structured performance tests. **GP Virtual Monitoring System** (3rd Year Project)

- Built a monitoring platform with real-time sensor data acquisition and a secure database.
- Performance-tested stable handling of multiple simultaneous inputs.

AI ADAS Detection System (Final Year Project)

- Designed a low-cost Advanced Driver Assistance System using computer vision, deploying real-time object detection on embedded hardware at the edge.
- Validated system performance through latency testing to ensure real-time responsiveness.

PROFESSIONAL EXPERIENCE

Cabin Crew, Ryanair | Aug 2026 – Oct 2026

- Carried out safety-critical duties, including pre-flight checks, passenger briefings and emergency procedures, to strict procedural standards.
- Delivered customer service in a fast-paced, time-sensitive environment as part of a multicultural crew.

Summer Analyst, Memon Transport Limited, Dublin | Jun 2020 – Jul 2021

- Maintained technical systems, managed cloud servers and resolved internal infrastructure issues.
- Supported employee onboarding and worked with senior leadership on operational planning.

LEADERSHIP AND VOLUNTEERING

Founder, Irish Sign Language Society, TU Dublin | Dec 2023 – Dec 2025

- Founded and scaled a student society to 50+ members, coordinating interpreters and logistics for accessible, inclusive events.

Student Brand Ambassador, Amazon (Prime), Deliveroo and TU Dublin | 2024 – 2025

- Nominated Student Ambassador of the Year (2024, 2025); elected Class Representative linking students and staff.

Events Officer, Treasurer and Secretary, Islamic Society, TU Dublin

- Planned student events, managed budgets and Charity Week fundraising, and kept meeting records.